

**1**  
17 rain gauges  
1,761 observed events



**2**  
PCA and  $k$ -means  
event classification



**3**  
Gaussian copula  
 $\sim 10^6$  synthetic events



**4**  
MaxDiss selection  
1,000 selected events



**8**  
Return-period depths  
at each cross-section



**7**  
Compound Poisson  
frequency model



**6**  
 $k$ -NN reconstruction  
over the full ensemble



**5**  
HEC-HMS + HEC-RAS 1D  
303 cross-sections